

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
 (An Autonomous Institution affiliated to VTU, Belagavi)  
**Third Semester B.E. (E&C) (Credit System) Degree Examinations**  
 Supplementary - April 2023

**20EC303 - ELECTRONIC DEVICES**

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

**Unit – I**

- |                                                                                              | Marks | BT* | CO* | PO* |
|----------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) With cross sectional diagram for each step, explain the fabrication of NPN transistor. | 12    | L*2 | 1   | 1   |
| b) With neat diagram, explain thermal oxidation process.                                     | 8     | L2  | 1   | 1   |
| 2. a) Explain the fabrication of PN junction diode with relevant diagrams.                   | 12    | L2  | 1   | 1   |
| b) Explain with necessary diagram silicon crystal growth method.                             | 8     | L2  | 1   | 1   |

**Unit – II**

- |                                                                                                                                                                                                                                                                                                                                                 |    |    |   |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 3. a) Derive the expression for built-in voltage of an unbiased PN Junction.                                                                                                                                                                                                                                                                    | 10 | L2 | 2 | 1 |
| b) Calculate (i) electron drift velocity (ii) time taken by an electron to cross the length of the bar (iii) resistivity of the Si uniform bar of n-type with 4 $\mu\text{m}$ length, cross sectional area of $0.25 \mu\text{m}^2$ and voltage of 1 V across it. Given $N_D = 10^{17}/\text{cm}^3$ and $\mu_n = 1350 \text{ cm}^2/\text{V-s}$ . | 6  | L3 | 2 | 1 |
| c) For the circuit shown in Fig. Q3(c), draw the output waveform if the input voltage is a 20 V peak-to-peak sinusoidal signal and $V = 5 \text{ V}$ . Assume ideal diode.                                                                                                                                                                      |    |    |   |   |

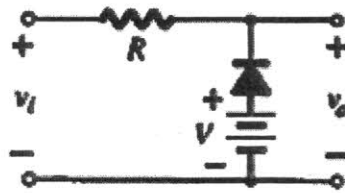


Fig. Q 3(c)

- |                                                                                                                                                   |    |    |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 4. a) Derive the expression for current in forward biased PN junction.                                                                            | 10 | L2 | 2 | 1 |
| b) For the circuit shown in Fig.Q4(c), draw the output waveform if the input voltage is 20 V peak-to-peak square wave signal. Assume ideal diode. |    |    |   |   |

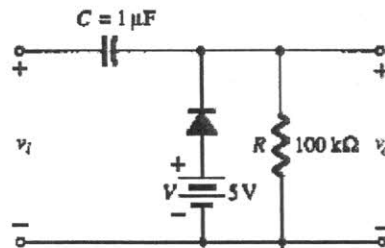


Fig.Q4(c)

- |                                                                                                                                                                                          |   |    |   |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| c) Calculate the built-in voltage of a PN junction with doping concentrations $N_A = 10^{18}/\text{cm}^3$ and $N_D = 10^{16}/\text{cm}^3$ . Given $n_i = 1.5 \times 10^{10}/\text{cm}^3$ | 6 | L3 | 2 | 1 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|

|  |   |    |   |   |
|--|---|----|---|---|
|  | 4 | L3 | 2 | 1 |
|--|---|----|---|---|

P.T.O.

## Unit – III

5. a) With relevant diagrams, explain the operation of NPN transistor in active mode and write the expression for terminal currents of it.
- b) Draw the small signal model of BJT neglecting early effect and derive the expression for small signal parameters.
- c) Consider the circuit shown in Fig.Q5(a) where  $V_1$  represents the signal generated by a microphone,  $I_S = 3 \times 10^{-16}$  A,  $\beta = 100$ , and Q1 operates in the active mode. (i) If  $V_1 = 0$ , determine the transconductance of Q1. (ii) If the microphone generates a 1-mV signal, how much change is observed in the collector current?

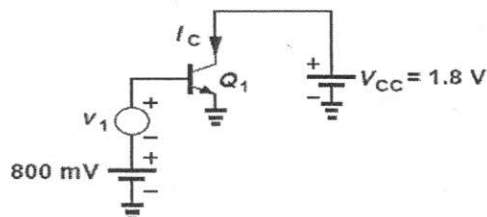


Fig.Q5(a)

6. a) Explain Early effect (or base width modulation) in BJT and draw the output characteristics of BJT in CE configuration with early effect. Also give the expression for collector current and output resistance.
- b) A transistor is biased at a collector current of 1 mA. Determine the small signal parameters of it if  $\beta = 100$  and  $V_A = 15$  V.
- c) With neat diagram explain the large signal model of NPN transistor.

## Unit – IV

7. a) Derive the expression for drain current of enhancement MOSFET.
- b) Draw the VI characteristics of MOSFET, mark various regions of operation and explain.
- c) Calculate the small-signal parameters of a MOSFET biased at a drain current of 0.5 mA if  $\mu_n C_{ox} = 100 \mu A/V^2$ ,  $W/L = 10$  and  $\lambda = 0.1 V^{-1}$ . Also draw small signal model.
8. a) Draw the high frequency model of a MOSFET and derive the expression for transit frequency of it.
- b) Explain body effect in MOSFET with relevant diagram.
- c) An NMOS device carries 1 mA with  $V_{GS} - V_{TH} = 0.6$  V and 1.6 mA with  $V_{GS} - V_{TH} = 0.8$  V. If the device operates in the triode region, calculate  $V_{DS}$  and  $W/L$ . Assume  $\mu_n C_{ox} = 100 \mu A/V^2$

## Unit – V

9. a) With cross sectional diagrams, explain the fabrication of GaAs MESFET.
- b) Explain the construction and working principle of LED. Also, list the applications of LED.
10. a) Draw the energy band diagram of MOS structure under thermal equilibrium, accumulation, depletion and inversion mode and explain.
- b) With neat diagram explain the working of solar cell and plot its VI characteristics clearly marking salient points.

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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